

Welcome to RASCAL!

ImageJ/Fiji Plugin User Manual

Version 1.8



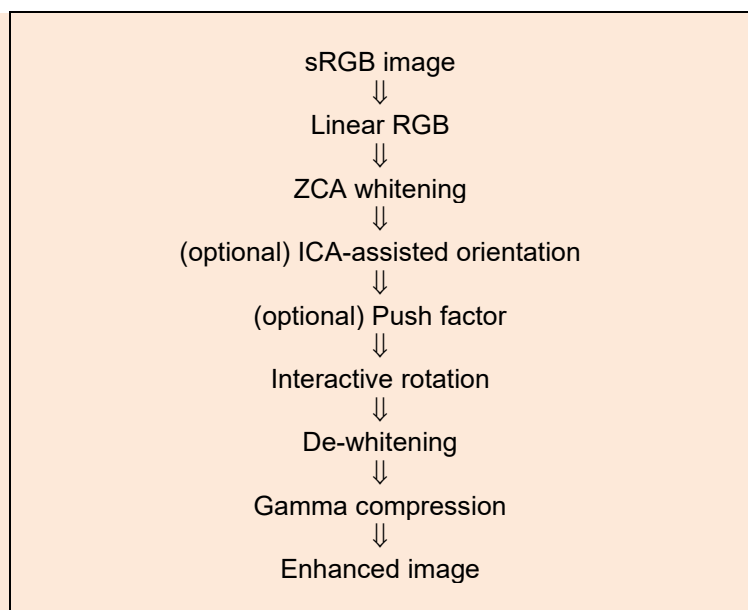
Edition	ImageJ / Fiji plugin
Platform	Windows / macOS / Linux

1. Presentation

RASCAL (**R**adial **A**ngular **S**tatistical **C**olor **A**daptation and **L**ifting) is an interactive software designed to reveal subtle chromatic variations in classical RGB digital images. The program was developed primarily for applications in cultural heritage and archaeological imaging, where faint color differences may correspond to pigments, surface alterations, or archaeological structures that are difficult to detect visually. The application performs a decorrelation of RGB color channels followed by an interactive exploration of color transformations. This approach allows the user to visually enhance color contrasts and to concentrate relevant chromatic information on one or several of the image channels. Unlike fully automated enhancement techniques, RASCAL relies on seamless interactive exploration. The operator visually evaluates the transformations and identifies the most informative configuration.

2. Conceptual overview

Digital images are typically encoded in the sRGB color space, where the three-color channels may be strongly correlated. Such correlations often confine color information to a restricted region of the RGB cube, which reduces the visibility of some chromatic variations. RASCAL addresses this limitation using the following workflow:



After whitening in the linear RGB color space, the covariance matrix of the color distribution becomes spherical, meaning that all chromatic directions have unit variance. Because such a distribution is invariant under orthogonal rotations, the user can rotate the color space to align chromatic variations of interest with the RGB axes. Note that the whitening transformation is computed once from the original image (or from a subset of it) and remains fixed during the entire exploration. ICA-assisted orientation can also be chosen as starting point, as it may help separate statistically independent chromatic components. In any case, subsequent rotations do not modify the statistical model. As a result, the rotation performed by RASCAL does not correspond to a simple rotation of the RGB axes. Instead, the rotation operates in the whitened color space, whose geometry depends on the covariance structure of the image. Consequently, the transformation is image-dependent and reflects the internal geometry of the color distribution. This process often reveals

structures that are difficult or impossible to perceive in the original image. After rotation, the transformed data are mapped back to the original RGB space through the inverse whitening (de-whitening) transform, restoring the original variance structure. The resulting linear RGB values are then converted back to sRGB through gamma compression for display.

The detailed mathematical description of the method is provided in the “Materials and Methods” section of the associated article.

3. Technical overview

The ImageJ/Fiji plugin is written in Java and compiled with Java 8 source and bytecode compatibility. The project is built with Maven and distributed in two variants: a thin JAR for Fiji, which relies on dependencies already bundled with that distribution, and a standalone fat JAR for standard ImageJ, produced using the maven-shade-plugin. The interactive preview is based on an OpenGL GLCanvas provided by JOGL and driven by an FPSAnimator at 60 fps, allowing rotations, zooming, and display updates to remain responsive. The color transformation matrices are prepared on the CPU, while the resulting matrix transformation is applied in real time on the GPU by a GLSL fragment shader during interaction. The user interface is built with Swing components such as JFrame, JSlider, JButton, and JToggleButton, and runs on the Event Dispatch Thread. Core dependencies include ImageJ 1.54 or later, JOGL, GlueGen, and Apache Commons Math.

4. Installation

The plugin requires ImageJ/Fiji 1.54 or later and an RGB color image to be open before launch. Fiji is recommended for most users because it provides a ready-to-use ImageJ distribution with many common components already included. Rascal GL is distributed as pre-built JAR files. For Fiji, copy Rascal_GL.jar into the plugins folder of your Fiji installation. For standalone ImageJ, copy Rascal_GL-standalone.jar into the plugins folder of your ImageJ installation; the file may be renamed to Rascal_GL.jar for simplicity. After copying the appropriate JAR file, restart the application. The plugin will appear in the Plugins menu as **Rascal GL**. In most cases, no further configuration is required. If ImageJ or Fiji is not yet installed on your system, Fiji can be downloaded from <https://imagej.net/software/fiji/downloads>. Alternatively, the standard ImageJ distribution can be downloaded from <https://imagej.net/ij/> and installed following the instructions for your operating system. Rascal GL is compiled for Java 8 compatibility. Most recent Fiji distributions include a suitable Java runtime, whereas standalone ImageJ may require a compatible Java installation depending on the package used.

On some systems, particularly macOS, the operating system may warn that the application or downloaded file comes from an unidentified developer. If ImageJ or Fiji is blocked at launch, open **System Settings**, go to **Privacy & Security**, and use the **Open Anyway** option after verifying that the file was obtained from a trusted source.

5. Loading an image and starting the plugin

Rascal GL operates as a plugin within ImageJ or Fiji. It does not provide its own image loading interface: images must first be opened using the standard ImageJ/Fiji tools, for example with File → Open. The supported file formats therefore depend on the ImageJ/Fiji installation and its available import plugins. Before launching Rascal GL, an RGB color image must be open and active. The plugin is started from the Plugins menu by selecting **Rascal GL**. At startup, Rascal GL checks the currently active image. If no image is open, or if the image is not in RGB color mode, an error message is displayed and the plugin stops. If the image is valid, Rascal GL extracts the pixel data, converts them to linear RGB, and computes the initial ZCA whitening parameters. For large images, this initialization may take a few seconds. Once complete, the preview window opens, the OpenGL canvas becomes visible, and the processing controls are available. Rascal GL always processes the image that was active at the time the plugin was launched. To process another image, close the Rascal GL window, bring the desired image to the front in ImageJ or Fiji, and launch the plugin again.

TIP: Rascal GL generally gives better results on sufficiently bright images. Dark images may benefit from brightness, exposure, or contrast adjustment in ImageJ or Fiji before processing.



6. Interface description

The ImageJ plugin interface consists of two main parts: the OpenGL preview canvas, which occupies most of the window, and the control panel located below it.

A: Preview canvas: displays the processed image in real time.

B: Preset buttons.

C: Channel buttons: Color, R, G and B.

D: ROI, ICA and Pol. +/- buttons.

E: Reset, Random, Split, and Save buttons.

F: Rotation sliders: separate seek bars for X, Y and Z rotations

G: Variance restoration and Push factor sliders.

7. Rotation and navigation

The user can interactively explore the transformed color space either through the dedicated sliders or through direct mouse interaction on the preview canvas:

Left mouse drag → rotation around the Z axis

Right mouse drag → rotation around the Y axis

Mouse wheel → rotation around the X axis

The **Random** button (or pressing the **Ctrl + Shift + R** keys) launches an automatic sequence of random color space orientations. A new random orientation is generated every 3 seconds, and the X, Y and Z sliders are updated accordingly. The sequence continues until the user interacts with any control, including a slider, mouse drag, mouse wheel movement, or the Push and Var controls, at which point it stops immediately. Pressing Random again starts a new sequence.

Zooming is centered on the pointer position and can be performed using:

Ctrl + mouse wheel / ⌘ + mouse wheel on macOS

→ zoom in/out

The Reset button (or pressing the R key) restores all rotation angles to zero, returns the Push factor and Variance restoration controls to their default values, clears the zoom, and restores the current ZCA whitening state.

8. Polarity inversion

The **Pol. +/-** button reverses the signs of the three Euler rotation angles used in the ZCA color space. In practice, the current rotation angles X, Y, and Z are replaced by $-X$, $-Y$, and $-Z$, and the rotation matrix is immediately recomputed from these new values. This provides a quick way to explore the opposite polarity of the current color enhancement, often making features that appear light on a dark background appear darker on a lighter background, or conversely. The displayed angle values, the processed image, and the point positions in the sRGB cube and ZCA space are updated accordingly. The operation is non-destructive and can be applied again to return to the previous angular configuration.

9. Split view

The **Split** button (or pressing the **T** key) divides a 2D image panel into two comparison areas. The left side displays the original reference image, while the right side shows the current processed rendering, including the active ZCA rotation, variance restoration, or channel display mode. A white vertical divider marks the boundary between the two views. The divider can be dragged horizontally with the mouse to adjust the relative size of the original and processed areas. When the image is zoomed or panned, the split remains aligned with the visible image area, making local before/after comparison easier.

10. Independent component analysis (ICA).

ICA-assisted orientation is an optional automatic transformation of the whitened color space. It is computed after ZCA whitening by clicking the ICA button (or pressing the **I** key), using a random sample of up to 100,000 pixels. Once the ICA transformation has been applied, the ICA button appears pressed, indicating that the current view corresponds to the ICA solution. ICA does not replace manual exploration; rather, it provides an automatically computed configuration that may help separate statistically independent chromatic components. The user may then continue exploring the image manually using the rotation sliders or mouse controls. Any subsequent modification of the rotation, Push factor, Variance restoration, or other processing parameters automatically deactivates the ICA state and returns the ICA button to its unpressed appearance, indicating that the current configuration no longer corresponds to the ICA solution.

11. Channel visualization

Four display modes are available:

- Color: full processed RGB image.

- R: red-channel grayscale view.

- G: green-channel grayscale view.

B: blue-channel grayscale view.

When a mode is selected, the corresponding button appears pressed.

12. ROI (Region Of Interest) processing

Clicking the **ROI** button (or pressing the **O** key) enters painting mode. In this state, the user paints directly on the image with a semi-transparent yellow brush to define the area of interest. This allows color enhancement to focus on specific chromatic elements while excluding areas whose colors differ markedly from the region under investigation.

The ROI workflow is as follows:

- Click **ROI** to enter paint mode.
- Left-drag to paint the region of interest.
- Double-click to validate the ROI and recompute the ZCA transformation from the selected pixels.
- Click **ROI** again to exit paint mode or to cancel an active ROI.

When the ROI is validated, the plugin recomputes the ZCA whitening parameters from the selected pixels of the original image. Although these parameters are estimated only from the ROI, the resulting transformation is applied to the entire image. If the painted area is too small, the operation is cancelled and a message prompts the user to paint a larger area. Once validated, the yellow overlay disappears while the ROI-derived ZCA transformation remains active. If ICA is subsequently applied, it is computed using the ROI-derived whitening transformation.

13. Presets

Presets allow you to save and instantly recall transformation states while working on an image. When a new image is loaded, Preset 1 is automatically initialized with the original transformation state, making it easy to switch back and forth between the original view and processed states. To save the current state as a preset, click the **Preset** button (or press the **P** key). The plugin stores the current rotation values, push factor, channel mode, Variance Restore, and, if a ROI has been applied, the full ROI transformation. Up to ten presets can be saved at a time. Each saved preset appears as a numbered button (1–10). To recall a preset, click its number. The sliders and channel mode are immediately restored. If the preset includes a ROI transformation, the image is also updated accordingly. Click the **x** button to clear all stored presets at once, except the first one corresponding to the original image. Presets are also automatically cleared when a new image is loaded. Note that presets are temporary: they are not saved after closing the plugin or restarting ImageJ/Fiji.

14. Variance Restore

The **Var** slider controls how much of the original color variance is restored after whitening.

At 1.00 (default), the full inverse whitening transform is applied. The original variance structure is restored, producing the standard RASCAL rendering with more visually stable colors.

At 0.00, variance restoration is disabled. The decorrelated and variance-equalized color components are mapped directly for display, producing a stronger DStretch-like effect, with enhanced chromatic contrast and less natural colors.

Intermediate values provide a continuous transition between standard RASCAL rendering and a stronger decorrelation-stretch-like visualization.

15. Push factor

The **Push Factor** is controlled by a dedicated seek bar. The displayed label is updated continuously. Values above 1.0 amplify the transformed effect, whereas values below 1.0 attenuate it. The default value is 1.0. The Push factor can also be adjusted with the keyboard: **Arrow Up** increases the value, and **Arrow Down** decreases it.

16. Saving processed images

Pressing the **Apply and save** button applies the current transformation to the full-resolution image and generates a new processed image. The result is automatically opened in a new ImageJ window and simultaneously saved as a JPEG file.

By default, the output file is written to the same directory as the source image using the filename format:

```
Rascal_<original_name>_yyyyMMdd_HHmm.jpg,
```

where yyyyMMdd_HHmm corresponds to the date and time of export. A confirmation message is displayed in the ImageJ status bar when the operation completes successfully. If the source image is not associated with a file on disk, the output is saved in ImageJ's current working directory.

17. Limitations of the current ImageJ plugin version

Compared with the full desktop application, the ImageJ plugin documented here does not expose several features available in the desktop version, including the processing of 3D models, interactive 3D color-cloud visualization, radial push processing, export of multiple view panels, session save/load capabilities, and expert/basic mode switching. The present manual therefore focuses exclusively on the functionality available within the plugin. TIFF and RAW files can be processed provided that they are successfully opened by ImageJ. However, the plugin always operates on the RGB representation generated by ImageJ and does not directly process RAW sensor data or multi-channel image formats.

18. Quick reference

Action	How to do it
Load an image	Open an RGB image in ImageJ/Fiji, then select Plugins → Rascal GL.
Rotate X	Move the X slider or scroll the mouse wheel or press Q/D .
Rotate Y	Move the Y slider or right-drag on the preview canvas or press Z/S .
Rotate Z	Move the Z slider or left-drag on the preview canvas or press A/E .
Zoom	Hold Ctrl (or ⌘ on macOS) and scroll the mouse wheel.
Random rotation	Click Random . Ctrl (or ⌘ on macOS) + Shift + R can also be used. Any manual interaction immediately stops the sequence.
Compute ICA rotation	Click ICA or press I .
Adjust variance restoration	Move the Var slider.
Change push factor	Move the Push slider. Arrow Up increases the value and Arrow Down decreases it.
Invert polarity	Click Pol. +/-
Inspect one channel	Click C , R , G , or B .
Paint an ROI	Click ROI , left-drag to paint, then double-click to process.
Cancel ROI	Click ROI again or reset the parameters. O toggles ROI mode and Esc cancels it.
Reset all transformations	Click Reset or press R .
Store the current configuration	Click Save preset or press P .
Clear presets	Click x .
Save the processed image	Click Apply and save .

19. Final note

RASCAL is an exploratory visualization tool. Processed images should always be interpreted in the scientific, archaeological, or documentary context of the original data.

20. Contact

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